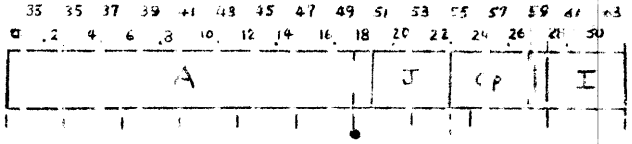
DIRECT INDEX

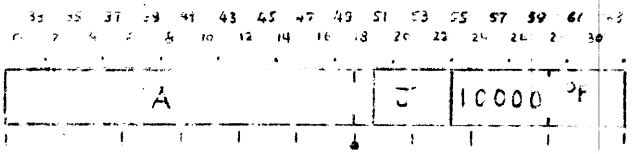
Op, J, A (I)



LX	LOAD INDEX
LV	LOAD VALUE
LC	LOAD COUNT
LR	LOAD REFIL.
KV	COMPARE VALUE
V+	ADD TO VALUE
V+C	ADD TO VAL & CNT
V+CR	ADD TO VAL, CNT & REF
SX	STORE INDEX
SV	STORE VALUE
SC	STORE COUNT
SR	STORE REFILL
KC	COMPARE COUNT
LVE	LOAD VAL EFFECTIVE
SVA	STORE VAL IN ADDR.
RNM	RENAME

IMMEDIATE INDEX

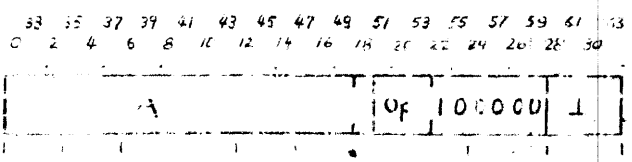
Op, J, A



C+I	ADD IMM. TO CNT
LVI	LOAD VAL IMM.
LCI	LOAD CNT IMM.
LRI	LOAD REF IMM.
KVI	COMP VAL IMM.
V+I	ADD IMM. TO VAL
V+IC	" " & CNT.
V+ICR	" " CNT & REF.
C-I	SUBT IMM FR CNT
LVNI	LOAD VAL NEG IMM.
KCI	COMP CNT IMM
LVS	LOAD VAL W/ SUM
KVNI	COMP VAL NEG IMM
V-I	SUBT IMM FR VAL
V-IC	" " & CNT
V-ICR	" " CNT & REF.

MISCELLANEOUS OPERATIONS

Op, A (I)



A

0 INITIATE STREAM

1 RESUME STREAM

R	REFILL
R ≠	REFILL ON COUNT ZERO
EX	EXECUTE
EXIC	EXECUTE INDIRECT & CNT
Z	STORE ZERO
(SDS)	BRKN DISJ STREAM
	CLEAR SMALL BLOCK MEM
	CLEAR LARGE BLOCK MEM

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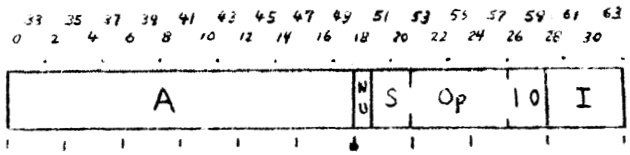
FLOATING POINT

Op (dd), A (I)

N - normalized
U - unnormalized

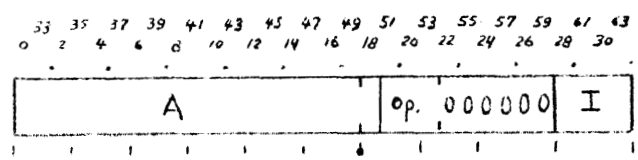


19	20		
0	x	SAME SIGN	S
1	x	ABSOLUTE SIGN	
x	0	SAME SIGN	
x	1	NEGATIVE SIGN	

0 0 0 0 0 1 0	+	ADD
0 0 0 0 0 1 1 0	L	LOAD
0 0 0 0 1 0	M+	ADD TO MEM
0 0 0 0 1 1	ST	STORE
0 0 0 1 0 0	K	COMPARE
0 0 0 1 0 1	KR	COMP FOR RANGE
0 0 0 1 1 0	*	MULTIPLY
0 0 0 1 1 1	/	DIVIDE
0 0 1 0 0 0	+MG	ADD TO MAGNITUDE
0 0 1 0 0 1	LWF	LOAD W/ FLAG
0 0 1 0 1 0	M+MG	ADD MAGN. TO MEM
0 0 1 0 1 1	SRD	STORE ROUNDED
0 0 1 1 0 0	KMG	COMP. MAGNITUDE
0 0 1 1 0 1	KMGR	COMP. MAGN. FOR RANGE
0 0 1 1 1 0	* +	MULTIPLY & ADD
0 0 1 1 1 1	RZ	RECIPROCAL DIVIDE
0 1 0 0 0 0	D+	ADD DOUBLE
0 1 0 0 0 1	DL	LOAD DOUBLE
0 1 0 0 1 0	LFT	LOAD FACTOR
0 1 0 0 1 1	SLO	STORE LOW ORDER
0 1 0 1 0 0	F+	ADD TO FRACTION
0 1 0 1 0 1	E+	ADD TO EXPONENT
0 1 0 1 1 0	D*	MULTIPLY DOUBLE
0 1 0 1 1 1	D/	DIVIDE DOUBLE
0 1 1 0 0 0	D+MG	ADD DOUBLE TO MAGN.
0 1 1 0 0 1	DLWF	LOAD DOUBLE W/ FLAG
0 1 1 0 1 0		
0 1 1 0 1 1	SRT	STORE ROOT
0 1 1 1 0 0	SHF	SHIFT FRACTION
0 1 1 1 0 1	E+I	ADD IMMED TO EXPONENT
0 1 1 1 1 0		
0 1 1 1 1 1		

UNCONDITIONAL BRANCHING

Op, A (I)



0 0 0 0 0 0 0 0 0	BE	BRANCH ENABLED
0 0 1 0 0 0 0 0 0	BD	BRANCH DISABLED
0 1 0	B	BRANCH
0 1 1	BR	BRANCH RELATIVE
1 0 0	BEW	BRANCH ENABLED & WAIT
0 1 0 1		BRANCH ENABLE INITIATE STRM
1 1 0 1		" " RESUME "
1 1 0	NOP	NO OPERATION

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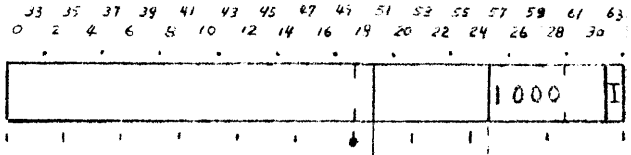
INDICATOR BRANCH

Op, A (K)

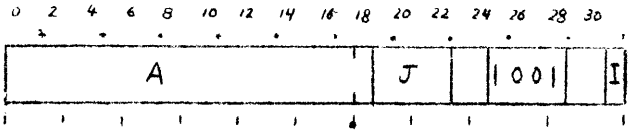


100000	BR ZERO BIT ~ LEAVE
000001	BR ONE BIT ~ LEAVE
000010	BR ZERO BIT - SET TO ZERO
000011	BR ONE BIT - SET TO ZERO
000000	MK MACH CHK INDIC
000001	IK INSTRUCTION CHK
000010	IJ INSTRUCTION REJECT
000011	EK EXCHANGE CONTROL CHK
000100	TS TIME SIGNAL
000101	CPUS OTHER CPU
000110	EKT EXCH CHK REJECT
000111	UNRJ UNIT NOT READY REJECT
001000	CBJ CHANNEL BUSY REJECT
001001	EPGK EXCH PROG CHK

etc.

COUNT AND BRANCH

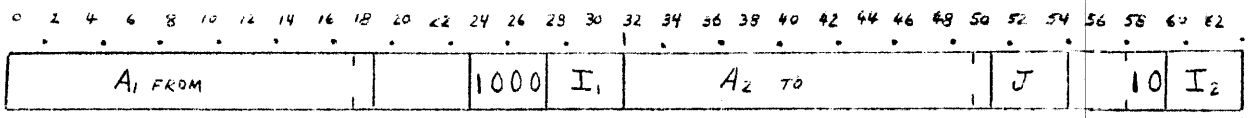
Op, J, A (K)



001000	CB CNT, BRAN IF NOT '0'
001001	CB= CNT, BRAN IF = 0
001010	CBR CNT, BRAN ≠0, REF=0
001011	CBR= CNT, BRAN IF=0, REF IF =0
000100	CBH +½ ADVANCE
000101	CB=H "
000110	CBRH "
000111	CB=RH "
100000	CB + +1 ADVANCE
100001	CB= + "
100010	CBR + "
100011	CB=R + "
110000	CB − DIMINISH
110001	CB= − "
110010	CBR − "
110011	CB=R − "

TRANSMIT

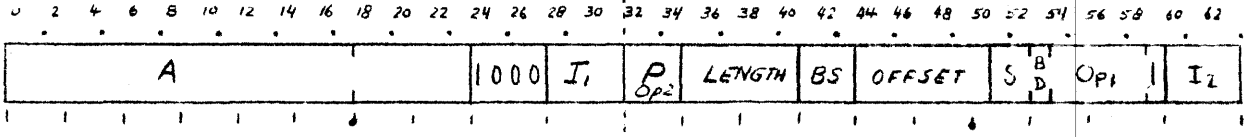
Op , J, A_1 (I_1), A_2 (I_2)



| | | |

INTEGER ARITHMETIC VFL

Op1 (Op2) (dd), A(I1), OFFSET(I2)



DIRECT ADDRESSING		0	0	0
ADD TO VALUE	V+I	0	0	1
ADD TO VAL & CNT	V+IC	0	1	0
ADD TO VAL, CNT, REF	V+ICR	0	1	1
IMMEDIATE MODE	I	1	0	0
SUB FR VAL	V-I	1	0	1
SUB FR VAL & CNT	V-IC	1	1	0
SUB FR VAL, CNT, REF	V-ICR	1	1	1

SIGNED ~ SAME	0	0
SIGNED ~ INVERT	0	1
UNSIGNED ~ PLUS	1	0
UNSIGNED ~ MINUS	1	1

0	0	0	0	0	1	+ ADD
0	0	0	0	1	1	LCV LOAD CONVERTED
0	0	0	1	0	1	L LOAD
0	0	0	1	1		CONNECT
0	0	1	0	0		M+ ADD TO MEM
0	0	1	0	1		LET LOAD FACTOR
0	0	1	1	0		ST STORE
0	0	1	1	1		CONNECT
0	1	0	0	0		K COMPARE
0	1	0	0	1		KE COMP IF EQUAL
0	1	0	1	0		KR COMP FOR RANGE
0	1	0	1	1		CONNECT
0	1	1	0	0		* MULTIPLY
0	1	1	0	1		CV CONVERT
0	1	1	1	0		/ DIVIDE
0	1	1	1	1		
1	0	0	0	0		+MG ADD TO MAGN
1	0	0	0	1		LTRCV LD TRANSIT CVRTD
1	0	0	1	0		LWF LOAD W/ FLAG
1	0	0	1	1		CONNECT
1	0	1	0	0		M+MG ADD MAGN TO MEM
1	0	1	0	1		M+1 ADD 1 TO MEM
1	0	1	1	0		SRD STORE ROUNDED
1	0	1	1	1		CONNECT
1	1	0	0	0		KF COMPARE FIELD
1	1	0	0	1		KFE COMP FLD IF =
1	1	0	1	0		KFR COMP FLD RANGE
1	1	0	1	1		CONNECT
1	1	1	0	0		*+ MULT & ADD
1	1	1	0	1		DCV CONVERT DOUBLE
1	1	1	1	0		LTRS LOAD TRANSIT & SET
1	1	1	1	1		

CONNECTIVE OPERATIONS

Op1 (Op2) (dd), A(I1), OFFSET(I2)

0	0	1	1	1	C CONNECT
0	1	1	1	1	CM CONNECT TO MEM
1	0				CT CONNECT FOR TEST
1	1				

0	0	0	0	0
0	0	0	0	1	m · a
0	0	0	1	0	m · ā
0	0	0	1	1	m
0	0	1	0	0	m̄ · a
0	0	1	0	1	a
0	0	1	1	0	m ∓ a
0	0	1	1	1	m ∨ a
0	1	0	0	0	m̄ · ā
0	1	0	0	1	m ≡ a
0	1	0	1	0	ā
0	1	0	1	1	m ∨ ā
0	1	1	0	0	m̄
0	1	1	0	1	m̄ ∨ a
0	1	1	1	0	m̄ ∨ ā
0	1	1	1	1	1

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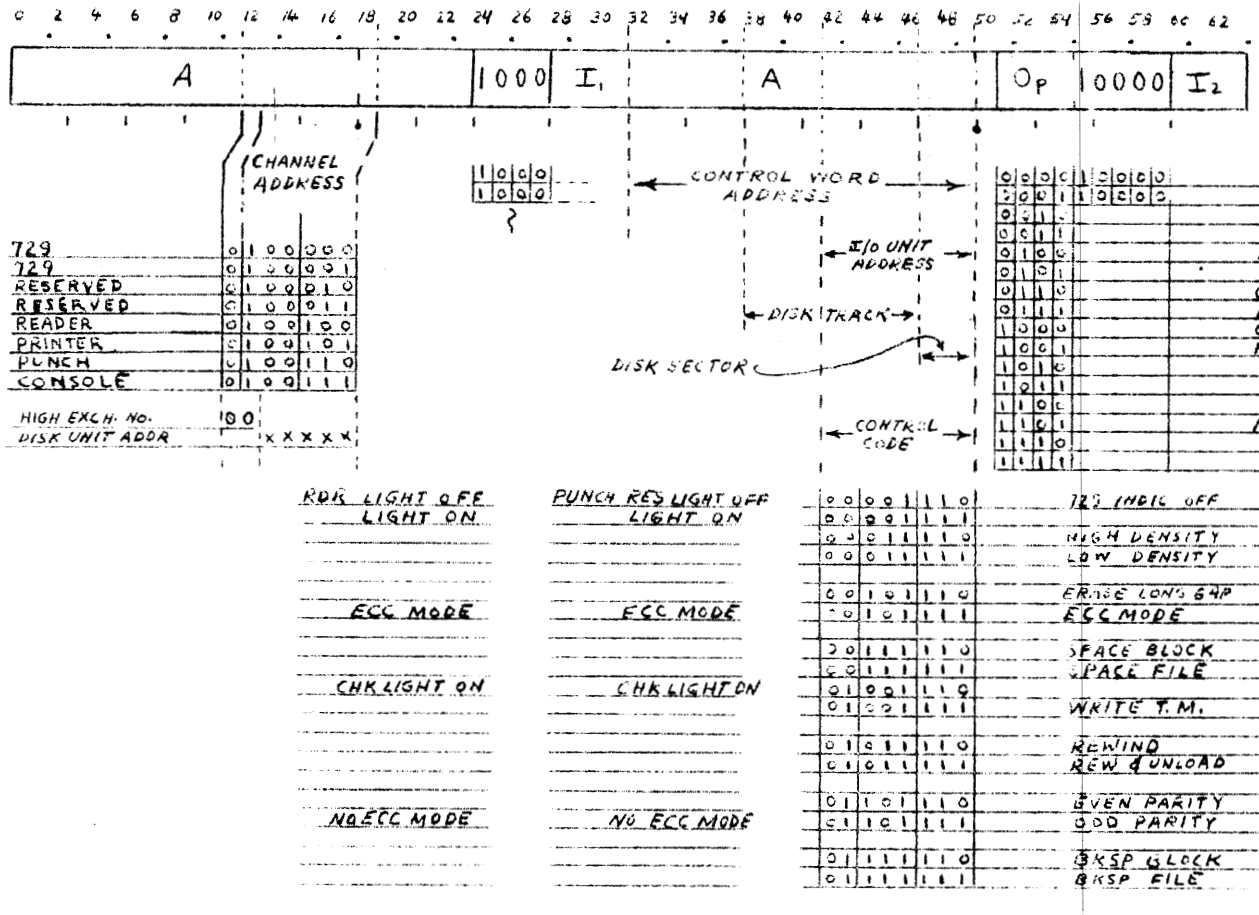
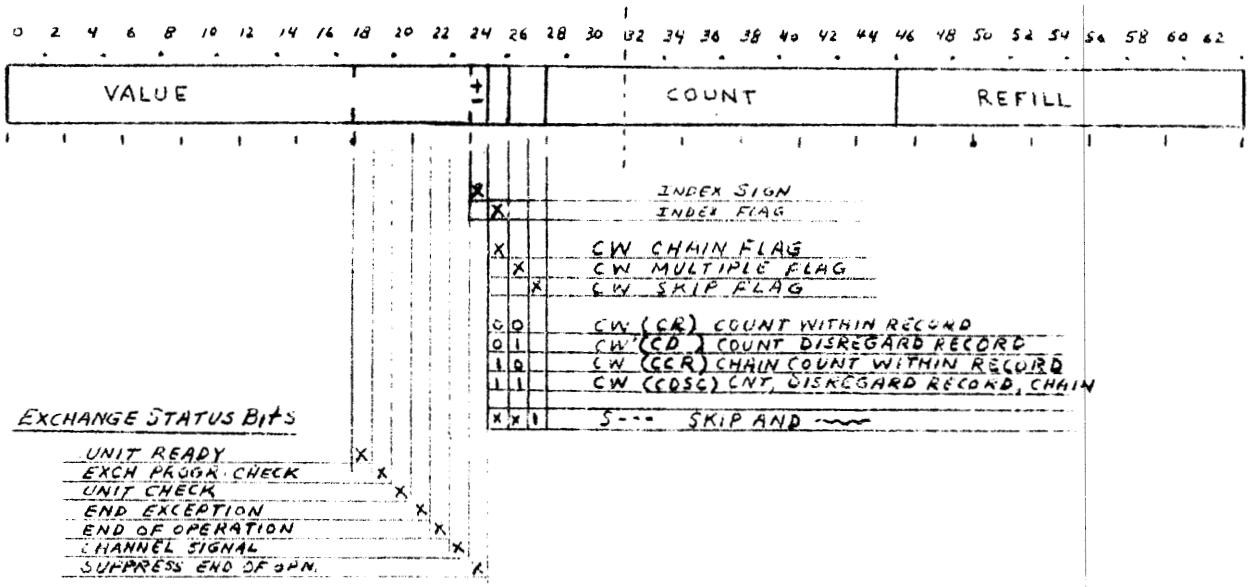
INPUT - OUTPUT

Op1 (Op2), C (I_1), A (I_2)



RDR LIGHT OFF LIGHT ON	PUNCH RES LIGHT OFF LIGHT ON		PRNT LIGHT OFF LIGHT ON
		729 INDIC OFF	
		HIGH DENSITY	
		LOW DENSITY	
		ERASE LONG GAP	
ECC MODE	ECC MODE	ECC MODE	
		SPACE BLOCK	
		SPACE FILE	
CHK LIGHT ON	CHK LIGHT ON		CHK LIGHT ON
		WRITE T.M.	
		REWIND	
		REW & UNLOAD	
		EVEN PARITY	
NO ECC MODE	NO ECC MODE	ODD PARITY	
		BKSP BLOCK	
		BKSP FILE	

INDEX AND/OR CONTROL WORD



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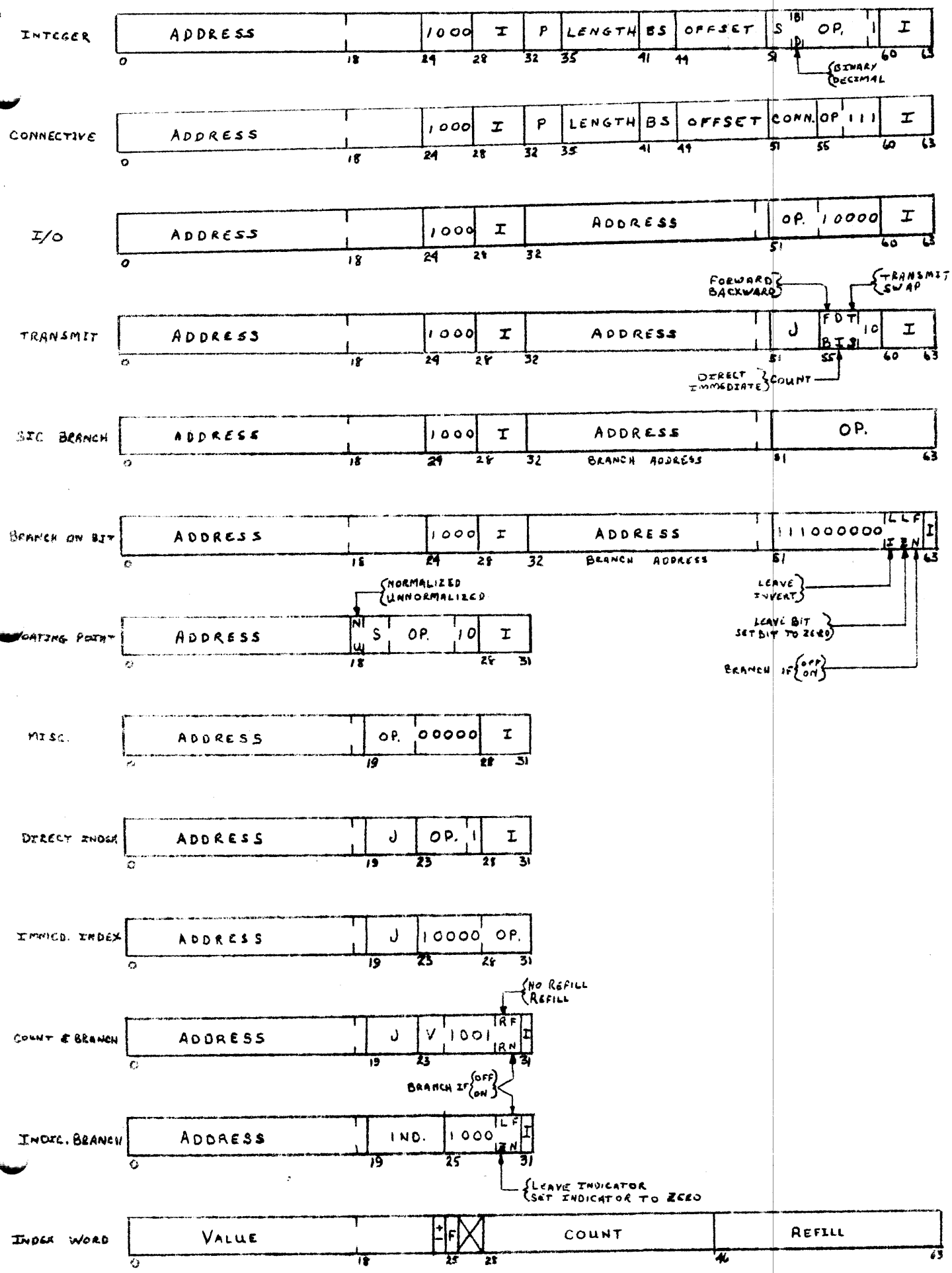
INTEGER
ADDRESS
1000
I
P
LENGTH
BS
OFFSET
S
OP.
I
0
18
24
28
32
35
41
44
51
60
63
BINARY
DECIMAL
CONNECTIVE
ADDRESS
1000
I
P
LENGTH
BS
OFFSET
CONN.
OP 111
I
0
18
24
28
32
35
41
44
51
55
60
63
I/O
ADDRESS
1000
I
ADDRESS
OP. 10000
I
0
18
24
28
32
51
60
63
FORWARD
BACKWARD
TRANSMIT
SWAP
TRANSMIT
ADDRESS
1000
I
ADDRESS
J
F D T
B I S
IC
I
0
18
24
28
32
51
55
60
63
DIRECT
IMMEDIATE
COUNT
SIC BRANCH
ADDRESS
1000
I
ADDRESS
OP.
0
18
24
28
32
BRANCH ADDRESS
51
63
BRANCH ON BIT
ADDRESS
1000
I
ADDRESS
11100000
L L F
I N
I
0
18
24
28
32
BRANCH ADDRESS
51
63
LEAVE
INVERT
LEAVE BIT
SET BIT TO ZERO
BRANCH IF
OFF
ON
NORMALIZED
UNNORMALIZED
FLOATING POINT
ADDRESS
N
U
S
OP.
10
I
0
18
25
31
MISC.
ADDRESS
OP.
00000
I
0
19
25
31
DIRECT INDEX
ADDRESS
J
OP.
1
I
0
19
23
25
31
IMMED. INDEX
ADDRESS
J
10000
OP.
0
19
23
25
31
NO REFILL
REFILL
COUNT & BRANCH
ADDRESS
J
V
1001
R F
R N
I
0
19
23
31
BRANCH IF
OFF
ON
INDIR. BRANCH
ADDRESS
IND.
1000
L F
I N
I
0
19
25
31
LEAVE INDICATOR
SET INDICATOR TO ZERO
INDEX WORD
VALUE
+
-
F
COUNT
REFILL
0
18
25
28
46
63

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